

How is AI bad for the environment? And how does it differ than let's say a search engine

Posted by u/CheeseheadTroy • 0 points • 8 comments

This is something I don't get. How is ai bad for the environment. And if it's for server maintenance and cooling and energy consumption how is it not any different than like googles search engine or any massive online service like gaming or anything like that....

This comes from a lot of talk about ai in gaming and everyone is talking about how it's bad for the environment.

I'm just trying to understand. This is in no way meant to be a political thing. I just genuinely don't understand it.

Comments

u/DiogenesKuon • 2 points • 2025-05-22 20:34 UTC

At current levels of usage, it really isn't. A single AI response uses around 10-15x what a single google response (prior to Google adding AI responses to searches). Which means if we start using AI for everything, or worse for a bunch of things we don't currently do, then the environmental problems could be problematic. That presumes we don't find more economical ways to use AI. DeepSeek has shown some pretty impressive results with significantly less compute power, and Google just released some early work on a diffusion based LLMs that looks like it could be more economical as well. Of course the complexity of tasks we give AIs could increase dramatically as well, leading to needing more power. So we will have to see what the long term environmental impact will be.

u/Primary-Nebula4397 • 1 points • 2025-05-22 20:08 UTC

So the difference between Google, game servers, and Ai generated stuff is the level of energy that's used.

Search engines are glorified databases, what they do is just bring to the surface what you've asked for. They aren't creating anything new, but instead just pulling from existing data. Game servers also aren't creating anything new they're just helping centralize all the data from across a series of networks. They still use up a huge amount of power for cooling, and thats why server rooms tend to be cold rooms, but nothing on the level of AI.

AI is creating entirely new information from your requests, in the sence that it's reading the existing info, analyzing for the necessary concepts, and then putting in tone of human speech. That requires

significantly more power, and to achieve cooling many companies are using water cooling strategies which aren't really good for the environment because of the heat that is still released into the air.

u/Delehal • 2 points • 2025-05-22 20:07 UTC

>And if it's for server maintenance and cooling and energy consumption how is it not any different than like googles search engine or any massive online service like gaming or anything like that

That's precisely the issue. It uses a lot more resources. It's not a little bit more. It's like comparing an electric scooter to a gas guzzling mega truck.

u/EdgySniper1 • 3 points • 2025-05-22 20:07 UTC

All of the things you list are in their own respect bad for the environment, but the main difference is scale. All things considered games, search engines, etc., don't take **too** much power - AI on the other hand can be such a massive power dump that for the past couple years tech companies have thrown ideas ranging from leasing power plant production to outright altering power grid infrastructure in order to reliably power AIs.

u/azuth89 • 5 points • 2025-05-22 20:02 UTC

Hosting and especially training AI models requires significantly more hardware than running a search engine, which in turn requires more manufacturing inputs, electricity and water.

That's the whole thing, one just needs more than the other.

u/CurtisLinithicum • 2 points • 2025-05-22 20:10 UTC

That's the perfect analogy:

Traditional db/algorithm/etc is pre-rendered graphics; AI is real-time rendering

u/PoopMobile9000 • 6 points • 2025-05-22 20:01 UTC

It uses more processing power

u/rabbithasacat • 5 points • 2025-05-22 20:05 UTC

And not just somewhat more resource intensive. **Wildly** more resource intensive.